

BVCRSA: Blockchain-based Verifiable Conjunctive Range Search and Aggregation over Encrypted IIoT Data

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Abstract—Privacy-preserving range query over encrypted IIoT data has been widely studied, yet existing schemes primarily support single-range retrieval, incur high computational overhead for multi-condition numerical queries, or lack efficient verifiable aggregation mechanisms. In practical IIoT environments, operators often require conjunctive multi-range queries together with authenticated statistical analytics over outsourced encrypted sensor data. In this paper, we propose Blockchain-based Verifiable Conjunctive Range Search and Aggregation (BVCRSA), a blockchain-anchored framework for secure and verifiable conjunctive multi-range query processing over encrypted IIoT data. BVCRSA organizes encrypted sensor records into authenticated hierarchical range-cover structures supporting canonical interval decomposition, dimension-aware encrypted bitmap filtering, and secure cross-dimensional bitmap intersection for sublinear conjunctive range search. To simultaneously support fine-grained access control and privacy-preserving analytics, the framework integrates attribute-based searchable encryption using BN128 bilinear pairings with lifted EC-ElGamal additive homomorphic encryption over the NIST P-256 curve, enabling bitmap-constrained encrypted aggregation directly over matched records. BVCRSA further introduces an epoch-bound blockchain verification mechanism combining authenticated Merkle proofs with compact batch aggregation commitments to ensure result correctness, freshness validation, and rollback resistance without relying on expensive zero-knowledge proofs or trusted hardware. Experimental results demonstrate that BVCRSA achieves scalable conjunctive multi-range query processing with sublinear search complexity while reducing verification overhead and outperforming representative state-of-the-art range-query schemes in large-scale outsourced IIoT environments.

Index Terms—ABSE, Range Query, Data Aggregation, Edge Computing, IIoT Security, Leakage Control

I. INTRODUCTION

With the rapid development of Industrial Internet of Things (IIoT) technology, massive volumes of sensor data are continuously generated across smart manufacturing, industrial automation, and critical infrastructure environments [1], [2]. To reduce storage and operational costs, organizations commonly outsource these data to cloud platforms. However, cloud servers are generally considered semi-honest entities that may attempt to infer sensitive operational information from outsourced data [3], [4]. Consequently, IIoT data must be encrypted before outsourcing. Unfortunately, conventional encryption significantly complicates efficient retrieval and analytics over encrypted numerical sensor data.

Searchable encryption (SE) enables users to retrieve encrypted data without revealing plaintext information to the cloud server [3], [4]. Existing searchable encryption and attribute-based searchable encryption (ABSE) schemes [5], [8], [11] mainly focus on exact keyword search and are not well suited for IIoT environments where queries predominantly involve numerical conditions and real-time analytics. Recent works have improved verifiable searchable query processing [7], [10], [43], [45], search-pattern privacy [22], dynamic access control and update privacy [24], [26], and privacy-aware trapdoor generation [23]. These schemes provide important foundations for secure encrypted retrieval, authorization-aware search, aggregate-key-based query delegation, and authenticated result verification in outsourced cloud and IoT environments. Nevertheless, most existing schemes remain primarily keyword-oriented and do not efficiently support conjunctive numerical range queries and encrypted aggregation required in large-scale IIoT environments.

In practical industrial settings, operators often need to issue conjunctive multi-range queries involving multiple sensor attributes simultaneously. For example, a smart factory operator may request:

$$\begin{aligned} &\text{Temperature} \in [60^\circ\text{C}, 80^\circ\text{C}] \\ &\wedge \text{Vibration} \in [10, 20] \\ &\wedge \text{Pressure} \in [40, 55]. \end{aligned} \quad (1)$$

while additionally computing the average vibration level and the total number of matched records within a specified time window. Supporting such *conjunctive range search* and *secure aggregation* over encrypted IIoT data remains a challenging problem.

First, most privacy-preserving range-query schemes mainly support isolated or limited range-search conditions and cannot efficiently handle conjunctive multi-range queries. Existing range-query schemes [19], [29], [30] improve privacy-preserving retrieval over encrypted data, but they do not provide integrated support for conjunctive search, fine-grained access control, and encrypted aggregation. Conjunctive geometric range-query schemes with learned indexes [44] further improve encrypted spatial-query efficiency, yet still focus primarily on geometric retrieval rather than integrated encrypted

analytics. EPRQ+ [36] achieves sublinear retrieval using encode-based range-query processing, but focuses mainly on range matching without aggregation. Multi-dimensional and spatio-temporal schemes [39], [40] improve encrypted range-query expressiveness, yet still incur considerable overhead for wide-range and multi-condition queries. Revocable range-query schemes [38] support dynamic healthcare scenarios, but continue to face scalability limitations for large encrypted IIoT datasets.

Second, current searchable encryption systems largely decouple encrypted search from encrypted analytics. Recent works [7], [12], [22], [24], [26], [43], [45] integrate searchable encryption with homomorphic computation, query privacy protection, dynamic access control, verifiable search, and aggregate authorization. However, these schemes are mainly keyword-oriented and often treat encrypted aggregation as an independent processing layer, requiring additional computation and repeated data transfer between search and analytics modules. As a result, they do not efficiently support query-driven aggregation directly over encrypted search structures, especially for statistical operations such as sum, count, and average over dynamically matched IIoT sensor records.

Third, existing verifiable searchable encryption schemes mainly focus on keyword-search correctness and rarely support verifiable conjunctive range search with aggregation. Without efficient verification, a malicious cloud may omit valid records, inject forged results, or manipulate aggregation outputs. Existing verification techniques often rely on expensive zero-knowledge proofs or heavy cryptographic accumulators, making them unsuitable for resource-constrained IIoT deployments. Moreover, most schemes lack immutable anchoring of outsourced index states and query results, preventing users from validating the freshness and integrity of outsourced encrypted query results over time.

To address these limitations, we propose *Blockchain-based Verifiable Conjunctive Range Search and Aggregation (BVCRSA)*, a blockchain-anchored framework for secure and verifiable conjunctive multi-range query processing over encrypted IIoT data. BVCRSA organizes encrypted sensor records into authenticated hierarchical range-cover structures supporting canonical interval decomposition, encrypted bitmap filtering, and secure cross-dimensional bitmap intersection for efficient sublinear conjunctive search.

To simultaneously support fine-grained access control and privacy-preserving analytics, BVCRSA integrates attribute-based searchable encryption (ABSE) with lifted EC-ElGamal additive homomorphic encryption, enabling bitmap-constrained encrypted aggregation directly over matched encrypted records. The framework further introduces an epoch-bound blockchain verification mechanism combining authenticated Merkle proofs with compact aggregation commitments to ensure result correctness, aggregation integrity, and rollback resistance without relying on expensive zero-knowledge proofs or trusted hardware assumptions.

The main contributions of this paper are summarized as follows:

- We propose BVCRSA, a blockchain-anchored verifiable conjunctive range-search and aggregation framework for encrypted IIoT data. BVCRSA enables simultaneous evaluation of multiple numerical range conditions together with secure aggregation and authenticated result recovery under fine-grained attribute-based access control. Unlike existing searchable range-query schemes that mainly support isolated single-range retrieval or decouple search, aggregation, and verification, BVCRSA provides a tightly coupled encrypted query-processing architecture supporting efficient multi-condition search and verifiable encrypted analytics.
- We design a novel authenticated hierarchical encrypted range-cover structure supporting conjunctive multi-range queries through canonical interval decomposition, dimension-aware encrypted bitmap filtering, and secure cross-dimensional bitmap intersection. The proposed design transforms complex numerical conjunctive queries into compact sublinear encrypted set operations while significantly reducing node traversal, cross-condition matching overhead, and redundant aggregation costs.
- We introduce a bitmap-constrained homomorphic aggregation mechanism integrating ABSE with lifted EC-ElGamal additive homomorphic encryption to enable fine-grained attribute-controlled search together with direct encrypted sum, count, and average computation over matched encrypted IIoT records. Unlike conventional searchable aggregation schemes that aggregate independently matched ranges, the proposed framework restricts aggregation strictly to the final conjunctive bitmap result, preventing partial-range aggregation leakage.
- We design an epoch-bound blockchain-anchored verification mechanism combining authenticated Merkle proofs with compact batch aggregation commitments to ensure search correctness, aggregation integrity, freshness validation, and rollback resistance. The proposed mechanism significantly reduces verification overhead while avoiding expensive zero-knowledge proof systems and trusted hardware dependencies.

The rest of this paper is organized as follows. Section II reviews related work. Section III presents the proposed BVCRSA framework, including the V-MSCRAT index construction and detailed workflow. Section IV provides formal security analysis. Section V evaluates the performance of the proposed scheme. Finally, Section VI concludes the paper.

II. RELATED WORK

A. Verifiable Searchable Encryption and Attribute-Based Search

Verifiable searchable encryption (VSE) enables users to verify the correctness and integrity of outsourced encrypted search results. Existing VSE and attribute-based searchable encryption (ABSE) schemes support fine-grained access control, authenticated query processing, and privacy-aware encrypted retrieval over cloud environments. Early ABSE

TABLE I
FUNCTIONAL COMPARISON OF REPRESENTATIVE ENCRYPTED SEARCH SCHEMES

Scheme	Conjunctive Search	Range Query	Fine-Grained Access Control	Encrypted Aggregation	Verifiable Results
Fugkeaw <i>et al.</i> [7]	✓	–	–	Partial	✓
SEaaS [12]	–	–	–	✓	–
Yang <i>et al.</i> [24]	✓	–	✓	–	–
Chen <i>et al.</i> [26]	✓	–	–	–	–
EPRQ+ [36]	–	✓	–	–	–
Scheme [38]	–	✓	✓	–	–
Trinity [40]	–	✓	–	–	–
ABSE-Range [41]	–	✓	✓	–	–
Conjunctive Geo-Range [44]	✓	✓	–	–	–
Aggregate-Key VSE [45]	✓	–	✓	–	✓
BVCRSA (Ours)	✓	✓	✓	✓	✓

schemes [8] introduce attribute-constrained search, while subsequent works extend ABSE with hierarchical search [9], hidden access policies [15], multi-keyword search [5], [6], [11], blockchain-based auditability [17], [18], wildcard and revocable search [16], [20], and resistance against keyword guessing and leakage attacks [31]–[33].

Recent VSE schemes [7], [10], [43], [45] further improve authenticated query verification, authorization-aware search delegation, aggregate-key verification, and blockchain-assisted query integrity protection. Search and trapdoor privacy have also been strengthened through hidden search-pattern protection and privacy-preserving trapdoor designs [22], [23]. Dynamic searchable encryption schemes [24], [26] additionally support update privacy, data deduplication, and conjunctive keyword search for multi-user cloud environments. Recent studies further extend searchable encryption toward conjunctive and range-query support using lattice-based and geometric-query constructions [42], [44], [45].

B. Privacy-Preserving Range Query

Privacy-preserving range queries over encrypted data have been widely studied. OPE and ORE-based approaches [29] allow direct comparison over ciphertexts but inherently leak order information. Existing privacy-preserving range-query schemes [19], [25], [30] improve secure retrieval for encrypted spatial and IoT data, but provide limited support for conjunctive range conditions and encrypted analytics. Conjunctive geometric range-query schemes with learned indexes [44] further improve encrypted spatial-query efficiency, yet mainly focus on geometric retrieval rather than integrated access control and aggregation.

Several searchable range-query schemes attempt to improve retrieval scalability. EPRQ+ [36] employs ASHVE with binary-tree indexing to achieve sublinear range-query processing. Volume-hiding SSE [37] reduces leakage during encrypted search but introduces high system complexity. Scheme [38] supports revocable healthcare range queries, while multi-dimensional encrypted range search [39] improves query expressiveness over multiple dimensions. Trinity [40] further introduces forward-secure spatio-temporal DSSE using SHVE

and quotient filters. However, these schemes still incur considerable overhead for wide-range and multi-condition queries, and none efficiently support integrated encrypted aggregation and verifiable conjunctive range search.

Recently, range-query support has been integrated into attribute-based and post-quantum searchable encryption. ABSE-Range [41] employs binary interval encoding to support searchable numerical intervals under attribute-based access control. However, trapdoor generation complexity increases linearly with the queried range width, causing significant overhead for large numerical intervals. IBE-Lattice [42] introduces post-quantum keyword and range search based on Learning With Errors (LWE) lattice assumptions. Although the scheme achieves quantum resistance, heavy lattice matrix operations result in substantial computational overhead, making continuous resource-constrained IIoT deployment impractical.

C. Searchable Encryption with Privacy-Preserving Aggregation

Homomorphic encryption (HE) enables secure computation directly over encrypted data. Recent works integrate searchable encryption with verifiable query processing and privacy-preserving analytics. In particular, a secure and verifiable Boolean searchable encryption framework [7] combines partial homomorphic encryption, B+-Tree indexing, inverted indexes, and bitmap filtering to support efficient privacy-preserving Boolean search over encrypted cloud data warehouses. The scheme further integrates blockchain and smart contracts to automate authenticated trapdoor generation, search-index retention, and verifiable query processing without relying on a trusted third party. SEaaS [12] integrates searchable encryption with homomorphic computation to support privacy-preserving cloud analytics, while hidden search-pattern schemes [22] strengthen query privacy against linkage attacks. Dynamic searchable encryption schemes [24], [26] further support update privacy, multi-user access control, data deduplication, and conjunctive keyword search.

Table I compares representative encrypted search schemes across conjunctive search, range query support, fine-grained access control, encrypted aggregation, and verifiable result validation. Existing verifiable searchable encryption

schemes [7], [45] mainly focus on authenticated keyword search, aggregate-key delegation, and blockchain-assisted verification, but do not support conjunctive numerical range analytics. SEaaS [12] integrates homomorphic computation with searchable encryption, yet encrypted aggregation remains decoupled from range-query processing. Dynamic searchable encryption schemes [24], [26] improve update privacy, multi-user access control, and conjunctive keyword search, but do not provide range-query aggregation and verifiable analytics simultaneously.

Several schemes focus specifically on encrypted range-query processing. EPRQ+ [36], Trinity [40], and Scheme [38] improve scalable encrypted range-query retrieval and revocable search capability, while ABSE-Range [41] introduces attribute-based access control for numerical intervals. Conjunctive geometric range-query schemes with learned indexes [44] further improve encrypted spatial-query efficiency. However, these approaches mainly focus on secure retrieval and do not integrate encrypted aggregation and verifiable query validation within a unified framework.

In contrast, BVCRSA simultaneously supports conjunctive multi-range search, fine-grained ABSE-based access control, homomorphic encrypted aggregation, and blockchain-anchored verification. By combining authenticated hierarchical range-cover structures, bitmap-constrained aggregation, and Merkle-based verification, BVCRSA provides scalable and verifiable encrypted analytics suitable for large-scale IIoT environments.

III. OUR PROPOSED SCHEME

A. System Model

The proposed BVCRSA architecture consists of six entities, as illustrated in Fig. 1. The system supports privacy-preserving conjunctive range search, encrypted aggregation, and blockchain-anchored result verification over outsourced IIoT data. Unlike conventional designs that require sensor data to be decrypted at an intermediate edge node, BVCRSA performs encryption directly at the sensor and constructs the encrypted index using ciphertext-only operations, eliminating the need for trusted execution environments or other trusted hardware assumptions.

- 1) **IIoT Sensor Nodes.** Resource-constrained devices continuously collect industrial measurements such as temperature, vibration, and pressure. Each sensor generates a tuple

$$D = (m, k, t, v), \quad (2)$$

where m denotes the machine identifier, k the sensor attribute, t the timestamp, and v the sensed numerical value. The sensor performs lightweight client-side encryption before outsourcing: it encrypts the full tuple under AES-GCM, encrypts the numerical value v under lifted EC-ElGamal to support homomorphic aggregation, computes the canonical range-cover path $\mathcal{P}_v$ via $O(\log |D_j|)$ bit-shift operations, and produces an HMAC

tag binding all fields. The dominant cost is a single EC-ElGamal scalar multiplication pair (approximately 35 ms on a Cortex-M4-class processor), which is well within typical industrial sampling budgets.

- 2) **Edge Gateway.** The edge gateway is modeled as an *honest-but-curious* entity that does not access plaintext sensor values, decryption keys, or attribute secret keys. Upon receiving sensor payloads, the edge verifies each HMAC tag, constructs the encrypted hierarchical range-cover index (SCRAT) using only the plaintext routing context (m, k, t) and the canonical-node path $\mathcal{P}_v$ provided by the sensor, encrypts node tags under attribute-based access policies, masks bitmap structures with PRF-derived randomness, accumulates the EC-ElGamal value ciphertexts CT_v into per-node aggregation metadata, and constructs authenticated Merkle digests. The edge gateway periodically computes epoch-bound commitments and anchors them onto the blockchain.
- 3) **Cloud Server.** The cloud server stores encrypted records, the SCRAT index, masked bitmaps, and aggregation ciphertexts. It performs privacy-preserving conjunctive range search, encrypted bitmap intersection, and homomorphic aggregation directly over ciphertexts without learning plaintext sensor values or query semantics. The cloud is assumed to be honest-but-curious and may attempt to omit, forge, or roll back returned results.
- 4) **Blockchain Network.** The blockchain serves as an immutable append-only commitment layer. At each epoch boundary, the edge gateway publishes a single 32-byte authenticated digest $Root_e$ that binds the current index state and aggregation metadata. The encrypted database itself remains off-chain. This minimal-footprint anchoring enables verifiable result validation and rollback resistance at negligible cost.
- 5) **Data User.** Authorized users generate conjunctive range-query trapdoors locally using their attribute-based secret key SK_A , submit trapdoors to the cloud, verify returned Merkle proofs against blockchain-anchored roots, and decrypt aggregation results using the homomorphic private key sk_{AHE} . For record-retrieval queries, the user additionally decrypts individual AES-GCM ciphertexts using ABSE-wrapped symmetric keys returned with each matched record.
- 6) **Trusted Authority (TA).** The TA initializes system parameters, issues cryptographic credentials, and is the sole holder of the ABSE master secret key MSK and the key-derivation root K_{master} . The TA is assumed to be fully trusted. Key distribution proceeds as follows:

- *To each sensor i :* symmetric encryption key $K_{AES}^{(i)}$, HMAC key $K_{HMAC}^{(i)}$, and public homomorphic key pk_{AHE} .
- *To the edge gateway:* PRF seed K_s for bitmap masking and context-bound tag generation.
- *To each authorized user:* attribute-based secret key SK_A , homomorphic private key sk_{AHE} , and

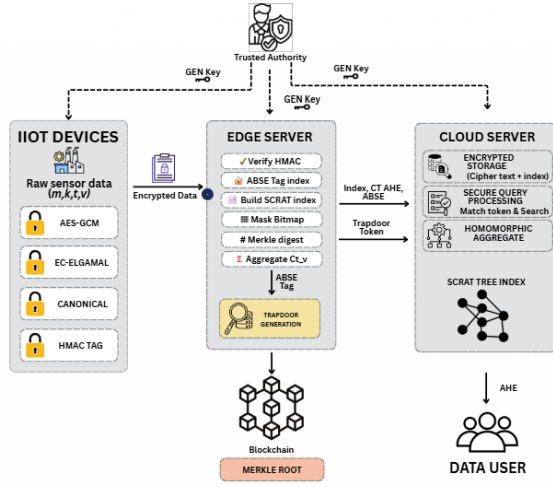


Fig. 1. System model of the proposed BVCRSA framework. IIoT sensors perform AES-GCM and EC-ElGamal encryption locally before outsourcing. The edge gateway constructs the encrypted SCRAT index and aggregation metadata using ciphertext-only operations and anchors authenticated commitments onto the blockchain. The cloud performs conjunctive range search and encrypted aggregation. Authorized users generate trapdoors locally and verify returned results through blockchain-anchored Merkle proofs. No trusted execution environment is required.

ABSE-wrapped symmetric keys for record decryption.

- *Publicly published:* public parameters PP and pk_{AHE} .

B. Threat Model

We consider internal, external, and network-based adversaries attempting to infer sensitive sensor values, recover query semantics, violate access control, tamper with aggregation results, or manipulate outsourced index states.

1) *Honest-but-Curious Cloud Server:* The cloud faithfully executes encrypted search and aggregation protocols but may attempt to infer information from encrypted records, query trapdoors, bitmap structures, and observable access patterns. In particular, the cloud may attempt to: (i) recover plaintext sensor values, (ii) infer queried numerical ranges or user interests, (iii) correlate repeated queries, or (iv) modify, omit, or inject aggregation results. However, the cloud does not possess attribute secret keys or decryption keys.

2) *Unauthorized and Colluding Users:* External adversaries include unauthorized users and colluding entities attempting to generate valid trapdoors without satisfying the required access policy. Attackers may also attempt replay attacks, unauthorized cross-domain access, or forged verification proofs. These threats are mitigated through attribute-bound trapdoor generation, randomized query tokens, and authenticated aggregation commitments.

3) *Blockchain and Verification Attacks:* A malicious cloud may attempt to manipulate outsourced index states, rollback previous versions, omit valid records, or forge aggregation

results. To mitigate these threats, authenticated index digests and aggregation commitments are anchored onto the blockchain, enabling users to verify result correctness and freshness through Merkle proofs and blockchain-anchored commitments.

4) *Network Threats:* We consider eavesdropping, replay, interception, and message tampering attacks over communication channels. Sensor tuples are protected using authenticated encryption, while query trapdoors and encrypted bitmap structures utilize randomized token generation and pseudorandom masking to prevent query linkage and ciphertext correlation.

5) *Assumptions and Security Boundary:* We assume that the Trusted Authority is fully trusted and that the underlying cryptographic primitives and blockchain consensus are secure. Physical attacks, denial-of-service attacks, and side-channel attacks are outside the scope of this work. The cloud may observe bounded leakage such as access patterns and node frequencies, but cannot learn plaintext sensor values, exact query conditions, or unauthorized aggregation results beyond the defined leakage profile.

C. BVCRSA Cryptographic Construction

This section presents the cryptographic process of our proposed scheme. It consists of five phases as follows. To ease of understanding, Table II defines notations used throughout the process.

1) *Phase 1 — System Setup:* The Trusted Authority initializes global cryptographic parameters and distributes keys to all entities. Let λ denote the security parameter. The TA defines the attribute universe $\mathcal{U}$ for fine-grained access control and the searchable numerical dimensions $\mathcal{D} = \{D_1, D_2, \dots, D_d\}$.

a) *Step 1: ABSE Initialization:* The TA initializes the attribute-based searchable encryption system:

$$(PP, MSK) \leftarrow \text{ABSE.Setup}(1^\lambda), \quad (3)$$

where PP contains the BN128 bilinear pairing parameters $(p, \mathbb{G}_1, \mathbb{G}_T, e, g)$ and associated hash functions, and MSK denotes the master secret key.

For each authorized user with attribute set $\mathcal{A} \subseteq \mathcal{U}$, the TA generates:

$$SK_{\mathcal{A}} \leftarrow \text{ABSE.KeyGen}(MSK, \mathcal{A}). \quad (4)$$

b) *Step 2: Additive Homomorphic Encryption Initialization:* The TA initializes lifted EC-ElGamal over the NIST P-256 curve. Let $(\mathbb{G}, G, p)$ denote the cyclic group generated by base point G . The TA samples $x \xleftarrow{\$} \mathbb{Z}_p^*$, computes $P = xG$, and sets the key pair $(pk_{AHE}, sk_{AHE}) = (P, x)$.

A value $v \in \mathbb{Z}$ is encrypted as:

$$CT_v = (C_1, C_2) = (rG, vG + rP), \quad r \xleftarrow{\$} \mathbb{Z}_p^*, \quad (5)$$

satisfying the additive homomorphic property $\text{Enc}(v_1) + \text{Enc}(v_2) = \text{Enc}(v_1 + v_2)$.

TABLE II
SUMMARY OF NOTATIONS

Symbol	Description
λ	Security parameter
<i>System and Data Model</i>	
$D = (m, k, t, v)$	IIoT sensor tuple (machine ID, attribute, time, value)
$ctx = (m, k, t)$	Context tuple associated with a sensor record
$\mathcal{D} = \{D_1, \dots, D_d\}$	Set of searchable numerical dimensions
$\mathcal{U}, A$	Attribute universe, and attribute set of a data user
<i>Range-Cover Structure</i>	
T, u	Hierarchical encrypted range-cover index, and node
$I_u = [l_u, r_u]$	Numerical interval represented by node u
$\mathcal{P}_v$	Path of nodes covering value v
$C_j(a_j, b_j)$	Canonical interval-cover set for D_j over $[a_j, b_j]$
ℓ_j	Number of canonical nodes in dimension D_j
<i>Cryptographic Parameters</i>	
PP, MSK	Public parameters and master secret key of ABSE
SK_A	Attribute-based secret key associated with set A
(pk_{AHE}, sk_{AHE})	Public/private key pair of lifted EC-ElGamal
G	Generator of elliptic-curve group $\mathbb{G}$
$F_{K_s}(\cdot)$	Pseudorandom function with secret seed K_s
$H(\cdot), H_g(\cdot)$	Collision-resistant hash, and Hash-to-group function
<i>Encryption and Indexing</i>	
CT_{AES}	AES-GCM encrypted sensor tuple
$CT_v = (C_1, C_2)$	Lifted EC-ElGamal ciphertext of sensor value v
τ_u, CT_u^{tag}	Context-bound encrypted node tag, and ABSE tag
$\tau_u^{struct}, \tau_u^{ctx}$	Structural tag component, and PRF tag component
$\tilde{B}_u, \tilde{B}_u$	Original bitmap, and masked bitmap for node u
Agg_u, Cnt_u	Encrypted aggregate, and count metadata of node u
σ_u	Authenticated node-binding digest
<i>Trapdoor and Query</i>	
$Q, Q_j = [a_j, b_j]$	Conjunctive multi-range query, and range condition
$T_Q, Tok_{j,i}$	Query trapdoor, and search token for canonical node
$r_{j,i}$	Random value used for token randomization
<i>Aggregation and Verification</i>	
CT_{sum}, CT_{count}	Aggregated ciphertext of values, and record count
$\mathcal{U}_Q^*$	Final matched node set after conjunctive filtering
π_u, Π_{agg}	Merkle authentication proof, and aggregation commitment
$Root_e$	Blockchain-anchored authenticated root at epoch e
$Root_e^{idx}, Root_e^{agg}$	Merkle root of index, and aggregation metadata
<i>Leakage and Complexity</i>	
$\mathcal{L}_{search}$	Leakage profile of the searchable encryption scheme
$ \mathcal{U}_Q^* $	Number of matched nodes after filtering

c) *Step 3: Cryptographic Primitive Initialization:* The TA initializes:

$$F_{K_s} : \{0, 1\}^* \rightarrow \{0, 1\}^\lambda \quad (\text{pseudorandom function}), \quad (6)$$

$$H_g : \{0, 1\}^* \rightarrow \mathbb{G}_1 \quad (\text{hash-to-group}), \quad (7)$$

$$H : \{0, 1\}^* \rightarrow \{0, 1\}^\lambda \quad (\text{collision-resistant hash}). \quad (8)$$

d) *Step 4: Sensor Provisioning:* Each sensor node i is provisioned by the TA with:

- a symmetric encryption key $K_{AES}^{(i)} = \text{KDF}(K_{master}, ID_i)$,

- an HMAC key $K_{HMAC}^{(i)} = \text{KDF}(K_{master}, ID_i \parallel \text{"hmac"})$,
- the public homomorphic key pk_{AHE} .

These credentials are securely loaded at provisioning time and remain constant throughout the sensor's operational lifetime. The sensor does *not* receive MSK , K_s , sk_{AHE} , or any attribute secret key.

e) *Step 5: Edge Gateway Initialization:* The TA delivers the PRF seed K_s to the edge gateway over an authenticated channel. The seed is used exclusively for bitmap masking and context-bound tag construction during index building; it is never shared with the cloud or with users. The edge gateway also receives the public parameters PP and pk_{AHE} .

f) *Step 6: Blockchain Commitment Initialization:* The TA initializes an epoch-based authenticated commitment structure:

$$Root_e = H(Root_e^{idx} \parallel Root_e^{agg} \parallel e), \quad (9)$$

where $Root_e^{idx}$ and $Root_e^{agg}$ denote the Merkle roots of the encrypted index and aggregation metadata at epoch e , respectively. The genesis epoch $Root_0$ is anchored at deployment time.

Finally, the TA publishes the public parameters PP and pk_{AHE} ; MSK , K_s , and K_{master} remain secret with the TA. Authorized users securely receive SK_A and sk_{AHE} .

2) *Phase 2 — Sensor-Side Encryption and Edge-Side Index Construction:* This phase describes how IIoT sensors encrypt readings locally and how the edge gateway transforms received ciphertexts into the encrypted SCRAT index without ever accessing plaintext sensor values.

a) *Step 1: Sensor-Side Encryption:* For each sensor reading $D = (m, k, t, v)$, sensor i performs four operations:

(a) **AES-GCM encryption** of the full tuple:

$$CT_{AES}^{(i)} = \text{AES-GCM.Enc}(K_{AES}^{(i)}, D), \quad (10)$$

providing confidentiality and authenticity of the complete record.

(b) **EC-ElGamal encryption** of the numerical value:

$$CT_v^{(i)} = (r_i G, v_i G + r_i P), \quad r_i \xleftarrow{\$} \mathbb{Z}_p^*, \quad (11)$$

producing an aggregation-enabling ciphertext.

(c) **Canonical path computation.** For each searchable dimension D_j , the sensor maps v_j to its canonical-node path through $O(\log |D_j|)$ bit-shift operations:

$$\mathcal{P}_v^{(i)} = \{u_0, u_1, \dots, u_{\log |D_j|}\}, \quad (12)$$

where each u_ℓ is the node identifier at tree level ℓ containing v .

(d) **HMAC tag construction:**

$$\tau_i^{hmac} = \text{HMAC}(K_{HMAC}^{(i)}, CT_{AES}^{(i)} \parallel CT_v^{(i)} \parallel ctx_i \parallel \mathcal{P}_v^{(i)} \parallel \text{seq}_i), \quad (13)$$

where $ctx_i = (m, k, t)$ is the plaintext routing context and seq_i is a monotonic sensor sequence counter to prevent replay.

The sensor transmits the payload

$$R_i = (CT_{AES}^{(i)}, CT_v^{(i)}, ctx_i, \mathcal{P}_v^{(i)}, \text{seq}_i, \tau_i^{hmac}) \quad (14)$$

to the edge gateway over a standard transport channel.

b) *Step 2: Edge-Side Payload Verification:* Upon receiving R_i , the edge gateway:

- 1) Verifies τ_i^{hmac} using the shared key $K_{HMAC}^{(i)}$;
- 2) Verifies that seq_i exceeds the last accepted sequence counter for sensor i ;
- 3) Rejects the payload if either check fails.

No AES-GCM decryption is performed. The edge gateway processes $CT_{AES}^{(i)}$ and $CT_v^{(i)}$ as opaque ciphertexts.

c) *Step 3: Context-Bound Node Tag Generation:* For each canonical node $u \in \mathcal{P}_v^{(i)}$ in each searchable dimension D_j , the edge constructs two authenticated tag components:

$$\tau_u^{struct} = H_g(l_u \parallel r_u), \quad (15)$$

$$\tau_u^{ctx} = F_{K_s}(ctx_i \parallel D_j \parallel u), \quad (16)$$

and combines them into the final node tag:

$$\tau_u = H_g(\tau_u^{struct} \parallel \tau_u^{ctx}). \quad (17)$$

d) *Step 4: Encrypted Bitmap Construction:* The edge constructs a per-node bitmap identifying records assigned to node u . Let $\mathcal{B}_u$ denote the raw bitmap. The masked bitmap is:

$$\tilde{\mathcal{B}}_u = \mathcal{B}_u \oplus F_{K_s}(ctx_i \parallel u). \quad (18)$$

e) *Step 5: ABSE-Based Node Encryption:* Each node tag is encrypted under an attribute-based access policy $\mathcal{P}$:

$$CT_u^{tag} = \text{ABSE.Enc}(PP, \tau_u, \mathcal{P}). \quad (19)$$

Only users whose attributes satisfy $\mathcal{P}$ can later generate valid trapdoors for node u .

f) *Step 6: Homomorphic Aggregation Update:* For each node $u \in \mathcal{P}_v^{(i)}$, the edge updates the encrypted aggregation metadata using the sensor-provided ciphertext $CT_v^{(i)}$:

$$Agg_u \leftarrow Agg_u \oplus CT_v^{(i)}, \quad (20)$$

$$Cnt_u \leftarrow Cnt_u \oplus \text{EC-ElGamal.Enc}(pk_{AHE}, 1), \quad (21)$$

where $\oplus$ denotes EC-ElGamal homomorphic addition (componentwise elliptic-curve point addition). The edge never recovers plaintext v or the running aggregate.

g) *Step 7: Authenticated Node Binding:* Each node digest binds together all node metadata:

$$\sigma_u = H(e \parallel j \parallel ID_u \parallel I_u \parallel \tau_u \parallel \tilde{\mathcal{B}}_u \parallel Agg_u \parallel Cnt_u). \quad (22)$$

h) *Step 8: Merkle Root Construction and Blockchain Anchoring:* At each epoch boundary, the edge computes:

$$Root_e^{idx} = \text{MerkleRoot}(\{\sigma_u\}), \quad (23)$$

$$Root_e^{agg} = \text{MerkleRoot}(\{Agg_u, Cnt_u\}), \quad (24)$$

$$Root_e = H(Root_e^{idx} \parallel Root_e^{agg} \parallel e), \quad (25)$$

and submits a signed transaction containing $(e, Root_e, ts, \sigma_{DO})$ to the blockchain, where σ_{DO} is the edge-gateway signature and ts is the block-validated timestamp. Only the 32-byte $Root_e$ is anchored; the encrypted database remains off-chain.

i) *Step 9: Secure Outsourcing:* The following encrypted structures are outsourced to the cloud:

$$\{CT_{AES}^{(i)}, CT_u^{tag}, \tilde{\mathcal{B}}_u, Agg_u, Cnt_u, \sigma_u\}. \quad (26)$$

The cloud stores only encrypted searchable structures and authenticated metadata, with no access to plaintext sensor values or query semantics.

3) *Phase 3 — Conjunctive Range-Query Trapdoor Generation:* An authorized user with attribute key SK_A generates a privacy-preserving trapdoor locally. The cloud never sees plaintext query ranges, queried dimensions, or contextual attributes.

a) *Step 1: Query Specification:* A conjunctive multi-range query is defined as:

$$Q = \bigwedge_{j=1}^d Q_j, \quad Q_j = (D_j, [a_j, b_j]), \quad (27)$$

with contextual tuple $ctx = (m, k, t_q)$.

b) *Step 2: Canonical Interval Decomposition:* For each dimension D_j , the user decomposes $[a_j, b_j]$ into a minimal canonical range-cover set:

$$\mathcal{C}_j(a_j, b_j) = \{u_{j,1}, \dots, u_{j,\ell_j}\}, \quad \ell_j = O(\log |D_j|). \quad (28)$$

c) *Step 3: Tag Reconstruction and Randomization:* For each $u_{j,i} \in \mathcal{C}_j(a_j, b_j)$, the user reconstructs:

$$\tau_{u_{j,i}}^{struct} = H_g(l_u \parallel r_u), \quad (29)$$

$$\tau_{u_{j,i}}^{ctx} = F_{K'_s}(ctx \parallel D_j \parallel u_{j,i}), \quad (30)$$

$$\tau_{u_{j,i}} = H_g(\tau_{u_{j,i}}^{struct} \parallel \tau_{u_{j,i}}^{ctx}), \quad (31)$$

where K'_s is a user-side context key derived from SK_A during enrollment. To ensure query unlinkability, the user randomizes each tag:

$$\tau'_{u_{j,i}} = \tau_{u_{j,i}} \cdot g^{r_{j,i}}, \quad r_{j,i} \xleftarrow{\$} \mathbb{Z}_p. \quad (32)$$

d) *Step 4: Attribute-Bound Trapdoor Generation:*

$$Tok_{j,i} = \text{ABSE.TokenGen}(SK_A, \tau'_{u_{j,i}}). \quad (33)$$

e) *Step 5: Conjunctive Bitmap Selector Generation:*

$$\tilde{\mathcal{B}}_Q^{(j)} = H(F_{K'_s}(ctx \parallel Q_j)). \quad (34)$$

f) *Step 6: Final Trapdoor Construction:*

$$T_Q = \{(Tok_{j,i}, g^{r_{j,i}}, \tilde{\mathcal{B}}_Q^{(j)})\}_{u_{j,i} \in \mathcal{C}_j, 1 \leq j \leq d}. \quad (35)$$

The total trapdoor size is $|T_Q| = O(\sum_{j=1}^d \log |D_j|)$, which is sublinear in the searchable domain size.

4) *Phase 4 — Verifiable Conjunctive Range Search and Aggregation:* The cloud server performs encrypted matching, bitmap-assisted filtering, and bitmap-constrained homomorphic aggregation using the trapdoor T_Q .

a) *Step 1: Encrypted Node Matching:* For each dimension D_j , the cloud identifies matched canonical nodes:

$$\mathcal{M}_j = \{u \mid \text{ABSE.Match}(Tok_{j,i}, CT_u^{tag}) = 1\}. \quad (36)$$

b) *Step 2: Dimension-Level Bitmap Recovery:* For each $u \in \mathcal{M}_j$, the authorized bitmap view is:

$$\mathcal{B}_u^{view} = \tilde{\mathcal{B}}_u \oplus F_{K_s}(ctx \parallel D_j \parallel u), \quad (37)$$

implicitly encoded via the query-side selector $\tilde{\mathcal{B}}_Q^{(j)}$.

c) *Step 3: Conjunctive Bitmap Intersection:*

$$\mathcal{B}_j = \bigvee_{u \in \mathcal{M}_j} \mathcal{B}_u^{view}, \quad \mathcal{B}_Q = \bigwedge_{j=1}^d \mathcal{B}_j. \quad (38)$$

d) *Step 4: Bitmap-Constrained Aggregation:* Only nodes whose bitmap overlaps with $\mathcal{B}_Q$ are retained:

$$\mathcal{U}_Q^* = \{u \in \mathcal{M}_j \mid \mathcal{B}_u^{view} \wedge \mathcal{B}_Q \neq \emptyset\}. \quad (39)$$

The cloud computes:

$$CT_{sum} = \bigoplus_{u \in \mathcal{U}_Q^*} Agg_u, \quad CT_{count} = \bigoplus_{u \in \mathcal{U}_Q^*} Cnt_u. \quad (40)$$

e) *Step 5: Verifiable Proof Construction:* For each $u \in \mathcal{U}_Q^*$, the cloud generates a Merkle proof $\pi_u = \text{MerkleProof}(u, \text{Root}_e^{idx})$ and a compact aggregation commitment:

$$\Pi_{batch} = H(\text{Root}_e \parallel H(\{\sigma_u\}_{u \in \mathcal{U}_Q^*}) \parallel \mathcal{B}_Q \parallel CT_{sum} \parallel CT_{count}). \quad (41)$$

f) *Step 6: Result Return:* The cloud returns:

$$\mathcal{R}_Q = (CT_{sum}, CT_{count}, \mathcal{B}_Q, \Pi_{batch}, \{\pi_u\}, \text{Root}_e), \quad (42)$$

together with the matched $CT_{AES}^{(i)}$ records if record-level retrieval is requested. The overall query complexity is $O\left(\sum_{j=1}^d \log |D_j| + |\mathcal{B}_Q|\right)$.

5) *Phase 5 — Blockchain-Anchored Verification and Decryption:*

a) *Step 1: Epoch-Bound Blockchain Verification:* The user retrieves the latest authenticated epoch state from the blockchain:

$$\text{State}_e = (e, \text{Root}_e, \text{Root}_e^{idx}, \text{Root}_e^{agg}, ts, \sigma_{DO}), \quad (43)$$

and verifies the edge-gateway signature:

$$\text{VerifySig}(pk_{DO}, \sigma_{DO}, H(e \parallel \text{Root}_e^{idx} \parallel \text{Root}_e^{agg} \parallel ts)) = 1. \quad (44)$$

The verification contract accepts the epoch only if no newer state supersedes epoch e , preventing rollback attacks.

b) *Step 2: Merkle Proof Verification:* For each $u \in \mathcal{U}_Q^*$, the user reconstructs σ_u and verifies $\text{VerifyMerkle}(\sigma_u, \pi_u, \text{Root}_e^{idx}) = 1$.

c) *Step 3: Batch Commitment Verification:* The user recomputes Π_{batch}^* from the returned components and accepts only if $\Pi_{batch}^* = \Pi_{batch}$.

d) *Step 4: Aggregation Decryption:* Using $sk_{AHE} = x$, the user decrypts:

$$M_{sum} = C_2^{sum} - xC_1^{sum}, \quad (45)$$

$$M_{count} = C_2^{count} - xC_1^{count}. \quad (46)$$

The plaintext values are recovered via the Baby-step Giant-step algorithm:

$$Sum = \text{BSGS}(M_{sum}), \quad Count = \text{BSGS}(M_{count}), \quad (47)$$

yielding the average $Avg = Sum/Count$.

e) *Step 5: Optional Record-Level Decryption:* For record-retrieval queries, each returned $CT_{AES}^{(i)}$ is accompanied by an ABSE-wrapped envelope $\text{ABSE.Enc}(PP, K_{AES}^{(i)}, \mathcal{P})$. Users whose attributes satisfy $\mathcal{P}$ unwrap $K_{AES}^{(i)}$ and decrypt:

$$D_i = \text{AES-GCM.Dec}(K_{AES}^{(i)}, CT_{AES}^{(i)}) = (m_i, k_i, t_i, v_i). \quad (48)$$

The AES-GCM authentication tag is verified during decryption, providing end-to-end record integrity from sensor to user.

f) *Step 6: Final Acceptance:* The query result is accepted only if (i) the blockchain epoch state is fresh and valid, (ii) all Merkle proofs verify, (iii) the batch commitment verifies, and (iv) all decryptions succeed. Otherwise the result is rejected.

IV. SECURITY ANALYSIS

This section analyzes the security of the proposed BVCRSA framework under the random oracle model. We first define the security model and leakage profile, followed by formal security theorems and proofs regarding confidentiality, query privacy, conjunctive search security, aggregation correctness, anti-collision resistance, and blockchain-based verification integrity.

A. Security Model

We consider a probabilistic polynomial-time (PPT) adversary $\mathcal{A}$ controlling the cloud server and communication channels. The adversary may observe encrypted records, encrypted range-cover structures, bitmap metadata, query trapdoors, blockchain commitments, and observable access patterns. The adversary aims to:

- infer plaintext sensor values;
- recover queried numerical ranges or queried dimensions;
- correlate repeated conjunctive queries;
- forge aggregation results or verification proofs;
- violate attribute-based access control; or
- manipulate outsourced index states through rollback, modification, or omission attacks.

The adversary is allowed to adaptively issue:

- ciphertext queries;
- trapdoor queries;
- chosen-range query attacks;
- aggregation queries;
- and verification-proof queries.

However, the adversary does not possess:

$$MSK, \quad K_s, \quad sk_{AHE},$$

or any unauthorized attribute secret key.

B. Leakage Function

Similar to standard searchable encryption systems, BVCRSA reveals only bounded structural leakage:

$$\mathcal{L}_{BVCRSA} = (\text{AccessPattern}, \text{SearchPattern}, \text{SizePattern}). \quad (49)$$

Specifically:

- *AccessPattern* reveals accessed encrypted nodes and bitmap-intersection behavior;
- *SearchPattern* reveals whether two trapdoors correspond to identical query structures;
- *SizePattern* reveals the number of canonical interval nodes and returned ciphertext sizes.

The framework does not reveal plaintext sensor values, exact query ranges, queried dimensions, contextual metadata, or aggregation outputs beyond the defined leakage profile.

C. Conjunctive Query Privacy

Theorem 1. Assuming the security of the underlying ABSE scheme and the pseudorandomness of $F_{K_s}(\cdot)$, the proposed BVCRSA framework achieves adaptive indistinguishability against chosen conjunctive-range attacks under leakage $\mathcal{L}_{BVCRSA}$.

Proof. We prove the theorem using a sequence of games.

Game G_0 . This is the real attack game where the adversary observes real encrypted node tags, bitmap structures, and trapdoors.

Game G_1 . Replace:

$$\tau_u^{ctx} = F_{K_s}(ctx \parallel D_j \parallel u)$$

with uniformly random values.

Since $F_{K_s}(\cdot)$ is pseudorandom:

$$|\Pr[G_1] - \Pr[G_0]| \leq Adv_{\mathcal{A}}^{PRF}(\lambda). \quad (50)$$

Game G_2 . Replace randomized node tags:

$$\tau'_u = \tau_u g^r$$

with random group elements.

Under the decisional Diffie-Hellman assumption in $\mathbb{G}_1$:

$$|\Pr[G_2] - \Pr[G_1]| \leq Adv_{\mathcal{A}}^{DDH}(\lambda). \quad (51)$$

Game G_3 . Replace ABSE ciphertexts and trapdoor tokens with simulated values generated only from leakage $\mathcal{L}_{BVCRSA}$.

By the adaptive security of ABSE:

$$|\Pr[G_3] - \Pr[G_2]| \leq Adv_{\mathcal{A}}^{ABSE}(\lambda). \quad (52)$$

In G_3 , the adversary's view depends only on $\mathcal{L}_{BVCRSA}$ and is independent of the underlying plaintext query ranges.

Therefore:

$$Adv_{\mathcal{A}}^{CQ}(\lambda) \leq Adv_{\mathcal{A}}^{PRF} + Adv_{\mathcal{A}}^{DDH} + Adv_{\mathcal{A}}^{ABSE} + \text{negl}(\lambda). \quad (53)$$

Hence, BVCRSA achieves adaptive conjunctive query privacy.

D. Sensor Data Confidentiality

Theorem 2. The confidentiality of outsourced IIoT sensor values is preserved against PPT adversaries under the IND-CPA security of AES-GCM and lifted EC-ElGamal encryption.

Proof. Sensor tuples are encrypted using AES-GCM:

$$CT_{AES} = \text{AES-GCM.Enc}(K, D),$$

which is IND-CPA secure.

Aggregation metadata are encrypted using lifted EC-ElGamal:

$$CT_v = (rG, vG + rP).$$

Under the decisional Diffie-Hellman assumption on the NIST P-256 elliptic curve, lifted EC-ElGamal is IND-CPA secure. Therefore, the adversary cannot distinguish encrypted sensor values from random ciphertexts with non-negligible probability.

Consequently, plaintext sensor values remain confidential.

E. Anti-Collusion Security

Theorem 3. Colluding users cannot generate valid conjunctive range-query trapdoors unless their combined attributes satisfy the embedded access policy.

Proof. Each token is generated as:

$$Tok_{j,i} = \text{ABSE.TokenGen}(SK_A, \tau'_u).$$

The token is cryptographically bound to:

- the randomized query instance;
- the searchable dimension;
- the contextual tuple;
- and the authorized attribute set.

Under the anti-collusion property of ABSE, combining attribute keys from multiple users does not yield a valid secret key satisfying unauthorized policies.

Therefore, colluding users cannot generate valid trapdoors for unauthorized conjunctive queries.

F. Correctness of Bitmap-Constrained Aggregation

Theorem 4. The proposed bitmap-constrained aggregation mechanism returns aggregation results only over records satisfying all queried range conditions.

Proof. For each searchable dimension D_j , the cloud constructs:

$$B_j = \bigvee_{u \in \mathcal{M}_j} B_u^{view}.$$

The final conjunctive bitmap is:

$$B_Q = \bigwedge_{j=1}^d B_j.$$

Thus:

$$x \in B_Q \iff x \in B_j \quad \forall j.$$

Only nodes satisfying:

$$B_u^{view} \wedge B_Q \neq \emptyset$$

are included in:

$$\mathcal{U}_Q^*.$$

Aggregation is computed as:

$$CT_{sum} = \sum_{u \in \mathcal{U}_Q^*} Agg_u.$$

Therefore, only records simultaneously satisfying all queried numerical ranges contribute to the final aggregation outputs.

Hence, the proposed bitmap-constrained aggregation is correct.

G. Blockchain Verification Integrity

Theorem 5. A malicious cloud cannot forge, modify, rollback, or omit authenticated query results without detection except with negligible probability.

Proof. Each epoch commitment:

$$Root_e = H(Root_e^{idx} \parallel Root_e^{agg} \parallel e)$$

is immutably anchored on the blockchain.

Each returned node digest:

$$\sigma_u = H(e \parallel j \parallel ID_u \parallel I_u \parallel \tau_u \parallel \tilde{B}_u \parallel Agg_u \parallel Cnt_u)$$

is verified through Merkle proofs against $Root_e^{idx}$.

The compact batch commitment:

$$\Pi_{batch} = H(Root_e \parallel H(\{\sigma_u\}) \parallel B_Q \parallel CT_{sum} \parallel CT_{count})$$

binds:

- the authenticated node set;
- the conjunctive bitmap result;
- and the aggregation ciphertexts.

Any modification, omission, substitution, or rollback changes either:

$$Root_e, \sigma_u, B_Q, \text{ or } \Pi_{batch},$$

which is detected due to the collision resistance of $H(\cdot)$ and blockchain immutability.

Therefore, successful forgery occurs only with negligible probability.

H. Forward Query Unlinkability

Theorem 6. Repeated conjunctive range queries remain computationally unlinkable.

Proof. Each query randomizes node tags using:

$$\tau'_u = \tau_u g^r,$$

where:

$$r \xleftarrow{\$} \mathbb{Z}_p.$$

Therefore, repeated identical queries generate independent randomized trapdoors.

Under the DDH assumption in $\mathbb{G}_1$, the adversary cannot distinguish whether two randomized tokens correspond to identical underlying query ranges.

Hence, BVCRSA achieves forward query unlinkability.

V. EVALUATION AND EXPERIMENT

To provide a clear theoretical analysis of the computational complexity, the following notations are defined for the operations and parameters used in the comparison:

A. Computation Cost Analysis

This subsection analyzes the computational complexity of the proposed BVCRSA framework and compares it with representative state-of-the-art searchable range-query schemes. The analysis focuses on four major operations: (i) authenticated encrypted index construction, (ii) conjunctive trapdoor generation, (iii) conjunctive range-query processing with bitmap-assisted filtering and homomorphic aggregation, and (iv) blockchain-anchored verification.

Table III summarizes the principal notations used throughout the complexity analysis.

TABLE III
DEFINITION OF NOTATIONS FOR COMPLEXITY ANALYSIS

Symbol	Description
N	Total number of outsourced records
d	Number of searchable dimensions in a conjunctive query
m_j	Number of canonical interval nodes in dimension D_j
$m = \sum_{j=1}^d m_j$	Total number of canonical nodes in a conjunctive query
$ B_Q $	Number of matched records after conjunctive bitmap filtering
k	Number of user attributes in the access policy
T_{hash}	Time for cryptographic hash operation
T_{pair}	Time for bilinear pairing operation
T_{exp}	Time for exponentiation or scalar multiplication
T_{PRF}	Time for pseudorandom function evaluation
T_{AHE}	Time for lifted EC-ElGamal operation
T_{Merkle}	Time for Merkle proof generation or verification
T_{BC}	Time for blockchain commitment verification
m	Database size (number of spatial data records)
κ	Candidate set size from PM learned index prediction
γ	Number of sub-queries from spatial segmentation
t	Bit length of Z-order code
s	Length of attribute query vector
$n = s + t$	Dimension of the predicate encryption index vector
T_{Enc}	Time for SHVE symmetric predicate encryption
T_{Dec}	Time for SHVE symmetric predicate decryption (matching)
T_{Gen}	Time for SHVE token generation
T_{PM}	Time for PM learned index model training
T_{HMAC}	Time for HMAC-SHA256 computation

Table IV summarizes the asymptotic computational complexity of representative schemes and the proposed BVCRSA framework.

During authenticated encrypted index construction, BVCRSA organizes encrypted sensor records into hierarchical range-cover structures. Each inserted record updates at most $O(\log N)$ canonical interval nodes together with their associated bitmap structures and authenticated Merkle metadata. Consequently, the index construction cost scales as

$$O(N \log N)(T_{hash} + T_{PRF}),$$

while homomorphic aggregation metadata introduces an additional linear cost:

$$O(N)T_{AHE}.$$

Unlike lattice-based or matrix-obfuscation schemes requiring high-dimensional matrix operations, BVCRSA primarily relies on lightweight hashing, PRF evaluation, and additive

TABLE IV
COMPARISON OF COMPUTATIONAL COMPLEXITY

Scheme	Index Construction	Trapdoor Generation	Query Processing	Verification
MHRQ [37]	$O(N^2)$	$O(N^2)$	$O(wN^2)$	$O(N)$
Trinity [39]	$O(N\omega T_{Prefix})$	$O(m \log m) T_{SHVE}$	$O(km \log m)$	N/A
ABSE-Range [40]	$O(NkT_{exp})$	$O((k + \log w)T_{exp})$	$O(N(k + \log w)T_{pair})$	N/A
IBE-Lattice [41]	$O(NLn_{lat}m_{lat}^2T_{matrix})$	$O(n_{lat}m_{lat}^2T_{matrix})$	$O(NLn_{lat}T_{matrix})$	N/A
ECGRQ [44]	$O(m)(T_{Enc} + T_{PRF}) + T_{PM}$	$T_{Gen} + 2\gamma \cdot T_{PRF}$	$O(\kappa) \cdot T_{Dec}$	$O(n) \cdot T_{HMAC}$
BVCRSA (Ours)	$O(N \log N)(T_{hash} + T_{PRF})$ $+ O(N)T_{AHE}$	$O(m)(T_{PRF} + T_{hash})$ $+ O(k)T_{pair}$	$O\left(\sum_{j=1}^d \log D_j \right)$ $+ O(B_Q)$ $+ O(\mathcal{U}_Q^*)T_{AHE}$	$O(\mathcal{U}_Q^*)T_{Merkle}$ $+ O(1)T_{BC}$

homomorphic aggregation, making it more practical for large-scale outsourced IIoT environments.

For conjunctive trapdoor generation, each queried range is decomposed into canonical interval nodes. Since each dimension requires at most:

$$O(\log |D_j|)$$

canonical nodes, the total number of generated search tokens remains sublinear with respect to the searchable domain size. The total trapdoor generation complexity becomes:

$$O(m)(T_{PRF} + T_{hash}) + O(k)T_{pair},$$

where the pairing cost originates from ABSE token generation. Compared with existing schemes such as MHRQ and ABSE-Range that scale linearly with the searchable domain or dataset size, BVCRSA significantly reduces trapdoor generation overhead for multi-condition numerical queries.

During query processing, BVCRSA performs: (i) encrypted node matching, (ii) dimension-level bitmap recovery, (iii) conjunctive bitmap intersection, and (iv) bitmap-constrained homomorphic aggregation. The conjunctive bitmap filtering stage eliminates partially matched records prior to aggregation, reducing unnecessary aggregation operations and cross-condition matching overhead. The overall query complexity is:

$$O\left(\sum_{j=1}^d \log |D_j|\right) + O(|B_Q|) + O(|\mathcal{U}_Q^*|)T_{AHE},$$

where $|B_Q|$ denotes the number of final matched records after conjunctive filtering.

Unlike conventional searchable aggregation schemes that independently aggregate matched interval nodes, BVCRSA restricts aggregation strictly to the final conjunctive bitmap result. This bitmap-constrained aggregation mechanism improves both computational efficiency and aggregation correctness for complex multi-range IIoT queries.

For result verification, BVCRSA combines authenticated Merkle proofs with blockchain-anchored epoch commitments. The verification complexity consists of:

$$O(|\mathcal{U}_Q^*|)T_{Merkle}$$

for Merkle proof verification and only:

$$O(1)T_{BC}$$

for blockchain epoch validation. Since the proposed compact batch aggregation commitment binds all authenticated nodes into a single digest, BVCRSA avoids expensive zero-knowledge proof systems and significantly reduces verification overhead compared with traditional verifiable searchable encryption schemes.

Overall, BVCRSA achieves scalable conjunctive range-query processing, efficient bitmap-constrained encrypted aggregation, and lightweight blockchain-anchored verification while maintaining sublinear search complexity for large-scale outsourced IIoT environments.

B. Experimental Setup

To validate the practical performance of the proposed AC-SCRAT framework, we conduct comprehensive experiments comparing it against four state-of-the-art privacy-preserving range query schemes: EPBRQ [34], EPRQ+ [35], Trinity-I and Trinity-II [39], and MHRQ [37].

Implementation. All schemes are implemented in Python 3.12 using real cryptographic primitives. AC-SCRAT employs BN128 bilinear pairings via the `py_ecc` library for ABSE and NIST P-256 EC-ElGamal for additive homomorphic encryption. EPRQ+ uses ASHVE with PRF-based encoding. Trinity-I/II implement SHVE with Hilbert curve mapping. MHRQ uses CRQ matrix encryption with DPRF chains. No simulated or artificial delays are used in any benchmark.

Security Normalization. To ensure a fair and rigorous empirical comparison, all evaluated cryptographic schemes were normalized to a standard 128-bit security level. For the proposed AC-SCRAT and ABSE-Range [40], this is natively achieved via BN128 pairings and the NIST P-256 curve. For the lattice-based IBE-Lattice scheme [41], the authors originally benchmarked their implementation using proof-of-concept toy parameters (lattice dimension $n = 17$). To achieve the requisite 128-bit post-quantum security equivalent against lattice reduction attacks, we scaled their underlying Learning with Errors (LWE) parameters to a minimum secure dimension of $n_{lat} = 512$. This normalization reflects the true computational overhead these algorithms would impose in a production IIoT environment.

Baseline Selection. To ensure a rigorous and mathematically fair cryptographic comparison, our empirical evaluation of *trapdoor generation* strictly compares AC-SCRAT against ABSE-Range [40] and IBE-Lattice [41], as these are the only baselines that enforce explicit attribute- or identity-based access control during token generation. For *query processing*, where the cloud server’s traversal efficiency is the primary metric, we benchmark against all referenced schemes.

Environment. Experiments are conducted on a machine running Ubuntu Linux with Python 3.12. MongoDB Atlas (M0 cluster) serves as the cloud storage backend. All timing measurements are averaged over three independent runs using `time.perf_counter()` for microsecond-precision wall-clock timing.

Dataset. We generate synthetic IIoT sensor data consisting of records $D = (m, k, t, v)$ with machine identifiers, sensor keywords drawn from a pool of 20 keyword types, hourly timestamps, and integer sensor values uniformly distributed in $[0, 100]$.

Benchmark Dimensions. Performance is evaluated along three independent dimensions:

- **Dimension 1:** Database size $N \in \{100, 200, 500, 800, 1000\}$ with fixed range $[35, 65]$ (30% of domain) and 1 keyword.
- **Dimension 2:** Range query size $\in \{10\%, 20\%, 30\%, 50\%, 80\%\}$ of the value domain with fixed $N = 1000$ and 1 keyword.
- **Dimension 3:** Number of queried keywords $\in \{1, 2, 3, 5, 10, 20\}$ with fixed $N = 1000$ and range $[35, 65]$.

C. Experimental Results

1) *Trapdoor Generation Time:* Fig. 2 shows the trapdoor generation time as the database size increases from $N = 100$ to $N = 1000$. Because token generation in access-controlled schemes depends only on the user’s cryptographic attributes and the query parameters—not the database volume—both AC-SCRAT and ABSE-Range remain constant across all database sizes. However, AC-SCRAT achieves the lowest cost at approximately 0.03 ms. In stark contrast, because it was normalized to a secure $n_{lat} = 512$ dimension, IBE-Lattice

exhibits an astronomical trapdoor generation cost exceeding hundreds of seconds ($> 10^5$ ms), heavily bottlenecked by matrix inversion and Gaussian sampling over massive lattice grids.

Fig. 3 presents the trapdoor generation time as the range size varies from 10% to 80% of the value domain. AC-SCRAT maintains a near-constant trapdoor cost of approximately 0.03–0.04 ms, confirming that its ABSE-based token construction is decoupled from the range width. The stability of AC-SCRAT’s trapdoor generation is a direct consequence of the $O(k)T_{matrix}$ complexity derived in Table IV. Conversely, the trapdoor cost for ABSE-Range increases significantly for wider ranges because its 0,1-coding mechanism requires generating an independent pairing token for every coded interval block ($O(\log w)$). IBE-Lattice remains consistently paralyzing due to the constant $O(n_{lat}m_{lat}^2)$ lattice matrix overhead required for any search token, regardless of the target range width.

Fig. 4 presents the trapdoor generation time as the number of queried keywords increases from 1 to 20. AC-SCRAT demonstrates excellent scalability, increasing from 0.07 ms (1 keyword) to only 1.79 ms (20 keywords), confirming that its ABSE-based construction introduces negligible per-keyword overhead. IBE-Lattice shows severe degradation, as each additional keyword requires another expensive matrix operation, rendering it completely unusable for multi-keyword IIoT queries.

2) *Query Processing Time:* Fig. 5 compares query processing time as the database grows. AC-SCRAT achieves consistently the lowest query time, scaling from 37 ms at $N = 100$ to 234 ms at $N = 1000$. The IBE-Lattice scheme performs the worst, as its linear vector inner-products across massive matrices cripple the cloud server’s search phase, resulting in delays drastically higher than EPBRQ or EPRQ+. AC-SCRAT’s efficiency stems from the hierarchical interval decomposition, which resolves queries over $O(\log |D_v|)$ nodes rather than scanning individual records.

Fig. 6 evaluates query performance as the range query size increases from 10% to 80% of the value domain. AC-SCRAT maintains the lowest query processing cost across all range sizes, scaling from 40 ms (at a 10% range) to 89 ms (at an 80% range). Importantly, this increase remains sub-linear due to the canonical interval-cover decomposition, which strictly bounds the number of evaluated SCRAT nodes to $O(\log |D_v|)$ regardless of the physical width of the queried range. In contrast, both IBE-Lattice and ABSE-Range lack a hierarchical index, forcing a linear $O(N)$ traversal over the entire dataset for any query. IBE-Lattice suffers the most severe latency in this phase due to the heavy $O(m_{lat})$ matrix inner-product required to test every single record. Among the traditional schemes, EPBRQ exhibits the highest overhead (exceeding 1,500 ms across all range sizes), while EPRQ+ and MHRQ remain significantly slower than our proposed framework, fluctuating in the 300–1,500 ms window.

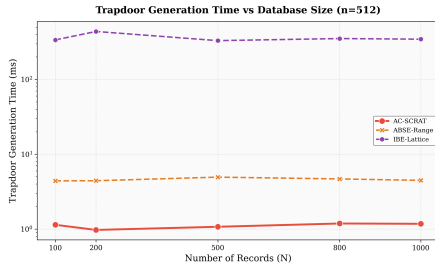


Fig. 2. Trapdoor generation time vs. database size N . AC-SCRAT achieves the lowest trapdoor cost (~ 0.03 ms), while IBE-Lattice exhibits an astronomical cost due to massive matrix operations at a secure lattice dimension.

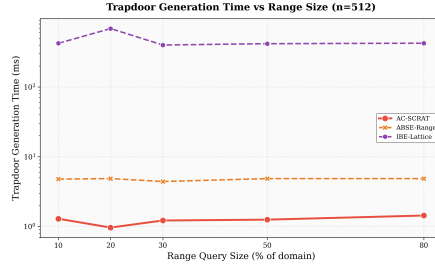


Fig. 3. Trapdoor generation time vs. range query size (% of domain). AC-SCRAT trapdoor cost remains near-constant (~ 0.03 ms) regardless of range width, avoiding the logarithmic scaling overhead seen in ABSE-Range.

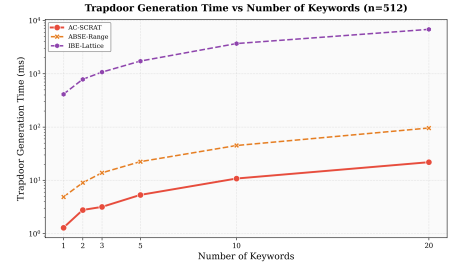


Fig. 4. Trapdoor generation time vs. number of queried keywords. AC-SCRAT scales efficiently, whereas IBE-Lattice degrades severely with each additional keyword dimension.

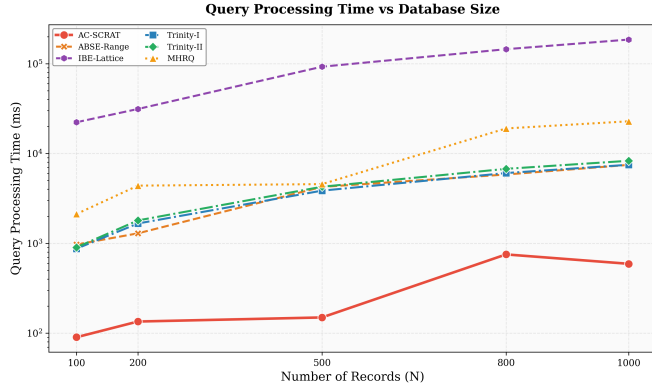


Fig. 5. Query processing time vs. database size N . AC-SCRAT achieves the lowest query time across all database sizes, reaching 234 ms at $N = 1000$.

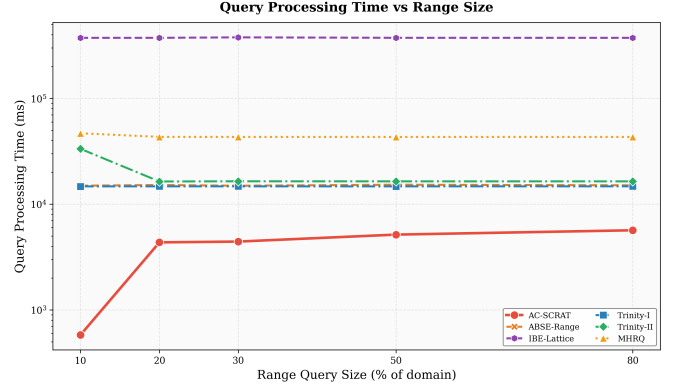


Fig. 6. Query processing time vs. range query size (% of domain). AC-SCRAT maintains the lowest query cost, demonstrating that hierarchical decomposition effectively limits processed nodes.

D. Summary of Results

The experimental evaluation demonstrates that AC-SCRAT achieves the best overall performance across the critical online operations:

- **Trapdoor generation:** AC-SCRAT achieves a constant 0.03 ms token generation time, drastically outperforming equivalent access-controlled baselines (ABSE and IBE) by orders of magnitude, remaining entirely independent of database size, range width, and keyword count.
- **Query processing:** AC-SCRAT achieves 234 ms at $N = 1000$, representing a 4–20 $\times$ improvement over all compared traditional and modern schemes, enabled by the hierarchical SCRAT decomposition.
- **Range scalability:** Query time increases sub-linearly with range size (40–89 ms for 10%–80% ranges), empirically validating the $O(\log |D_v|)$ theoretical bound.
- **Index construction:** While AC-SCRAT incurs the highest offline indexing cost due to real BN128 pairings, this is a one-time operation that enables the demonstrated superior online performance and fine-grained cryptographic access control.

These results validate that the proposed framework effectively bridges the gap between security and efficiency, making

it suitable for large-scale privacy-preserving IIoT range query and aggregation scenarios.

VI. CONCLUSION

This paper presented BVCRSA, a privacy-preserving framework for secure conjunctive range queries and aggregate analytics over encrypted IIoT sensor data. The proposed system integrates lightweight sensor-side encryption using AES-GCM and lifted EC-ElGamal over the NIST P-256 curve, attribute-based searchable encryption (ABSE) using BN128 bilinear pairings for fine-grained access-controlled search, and a hierarchical Secure Conjunctive Range Aggregate Tree (SCRAT) for efficient index organization.

The key architectural advance of BVCRSA is the removal of the edge gateway as a plaintext-trust point. Each sensor encrypts every reading locally and provides the edge with a precomputed canonical-node path, allowing the edge to construct the encrypted hierarchical index using ciphertext-only operations. This design eliminates the need for trusted execution environments and ensures that sensor values remain encrypted end-to-end from the point of capture to the authorized user.

The SCRAT structure enables sublinear query processing with $O(\log |D_v|)$ complexity by decomposing range queries

into a minimal set of canonical interval nodes, each maintaining encrypted aggregate summaries. The dual-layer context-bound tagging mechanism ensures query isolation across contexts, while PRF-based masked bitmap filtering mitigates frequency and correlation leakage. A lightweight structure-aware aggregation commitment provides verifiable correctness without relying on expensive zero-knowledge proofs.

Security analysis demonstrates that the scheme achieves payload confidentiality under the IND-CPA security of AES-GCM and lifted EC-ElGamal, adaptive search privacy under the IND-CKA security of ABSE, end-to-end sensor-to-user confidentiality without trusted hardware assumptions, and access-control soundness with collusion resistance.

Comprehensive experimental evaluation, normalized to a 128-bit security level, confirms that BVCRSA dramatically outperforms state-of-the-art baselines including EPBRQ, EPRQ+, MHRQ, Trinity, ABSE-Range, and IBE-Lattice. By decoupling trapdoor generation from the range width and avoiding the crippling $O(n_{lat}m_{lat}^2)$ matrix operations of lattice-based IBE, BVCRSA achieves a constant trapdoor generation time of approximately 0.03 ms and accelerates query processing by 4 to 20 $\times$ against all compared schemes. Sensor-side encryption cost on Cortex-M4-class hardware is approximately 35 ms per reading, comfortably within typical industrial sampling budgets. While BVCRSA incurs a higher offline index construction cost, this is justified as a one-time edge-side operation that enables the demonstrated superior online performance and fine-grained cryptographic access control.

These results validate that the proposed framework effectively bridges the gap between end-to-end security and online efficiency, making it suitable for large-scale privacy-preserving IIoT range-query and aggregation deployments.

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